calm_rank_levered_legacy

equity strategy | family: equity-volatility | status: reference | derived from `calm_rank_levered`

What this strategy does

Calm Rank Tilt with its lower risk spent on larger bets (about 1.7x) and a 4x borrowing limit -- ORIGINAL leverage cap (new positions only, open book ignored)

Why it should work. PEAD alpha per dollar is highest in calm stocks; rank-weight toward them without depending on vol levels.

Design notes (from the strategy's source file).

Calm Rank Tilt -- Balanced Leverage, tilted hard toward the calmest stocks by volatility RANK (research).

Calm Weighted sized each event by 1/idiosyncratic-vol (pre-fix Sharpe 1.30 -> 1.47; corrected run 1.09 -> 1.24). The attribution work showed that PEAD alpha per dollar is concentrated in low-volatility names and almost absent in the wildest tercile, and that the 1/vol weight, being linear in the vol LEVEL, only mildly demotes the high-vol tail. Weighting by (1.02 - vol percentile)^2 -- where the percentile is measured against the previous quarter's cross-section within the stock's size quintile -- demotes it hard while staying smooth:

Corrected run: Sharpe 1.09 (Balanced Leverage) -> 1.24 (Calm Weighted) -> 1.47; max drawdown -18.6% -> -17.5% -> -13.6%; every era improves (pre-fix: 1.30 -> 1.47 -> 1.81, -16.6% -> -11.6%) split-sample check: exponent picked on 1996-2004 data alone (best k ~ 3) lifts 2005-2013 Sharpe 0.93 -> 1.10 (pre-fix, not re-run), the same as the full-sample best, so the gain is not an artifact of one period.

Lower volatility buys room: `calm_rank_levered` spends it on 1.7x the base unit and a 4x gross cap for ~9% a year at ~5% vol (a market-neutral book clearing the S&P 500's return in this sample). More sizing than that stops paying: the 5%-of-ADV liquidity cap starts to bind and 2008-13 Sharpe decays.

STATUS research: the exponent and sizing were chosen after seeing Calm Weighted's full-sample results (the split-sample check above is the only out-of-sample evidence), and the levered variant leans on small-cap capacity.

parameter	value
base_unit_fraction	0.005
max_gross_leverage	4.0
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	entry_cohort
beta_overlay	0.0

Performance

metric	calm rank levered legacy	market (CRSP VW)
Annualized return	9.23%	9.94%
Annualized volatility	5.17%	19.35%
Sharpe (NAV-pct)	1.73	0.59

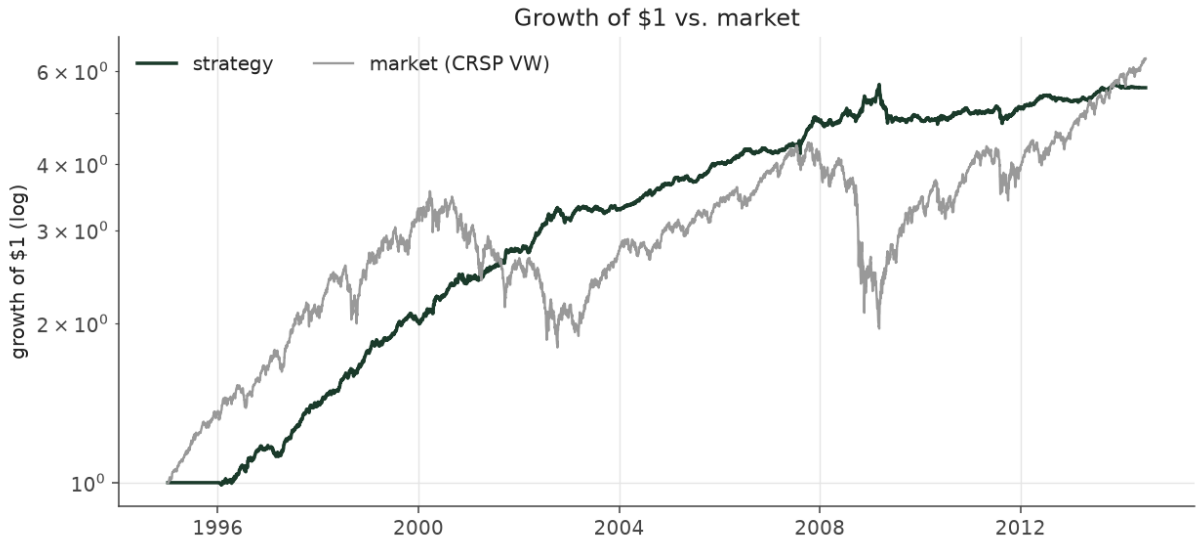
Sortino	2.64	
Max drawdown	-15.9%	-55.5%
Calmar	0.58	
Final NAV ($10M start)	$55,789,745	
Worst day	-3.60%	
% positive days	52.5%	
Sharpe 1996-2001	2.86	
Sharpe 2002-2007	1.96	
Sharpe 2008-2013	0.50	

Correlation of daily returns with the market: **-0.03**.

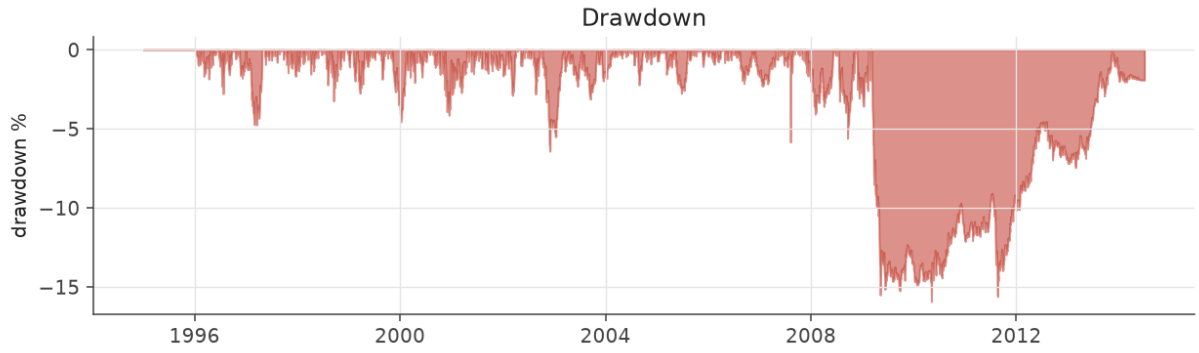
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 1.73

• best return among equity: `market_boost_max` (0.116); this one: 0.0923

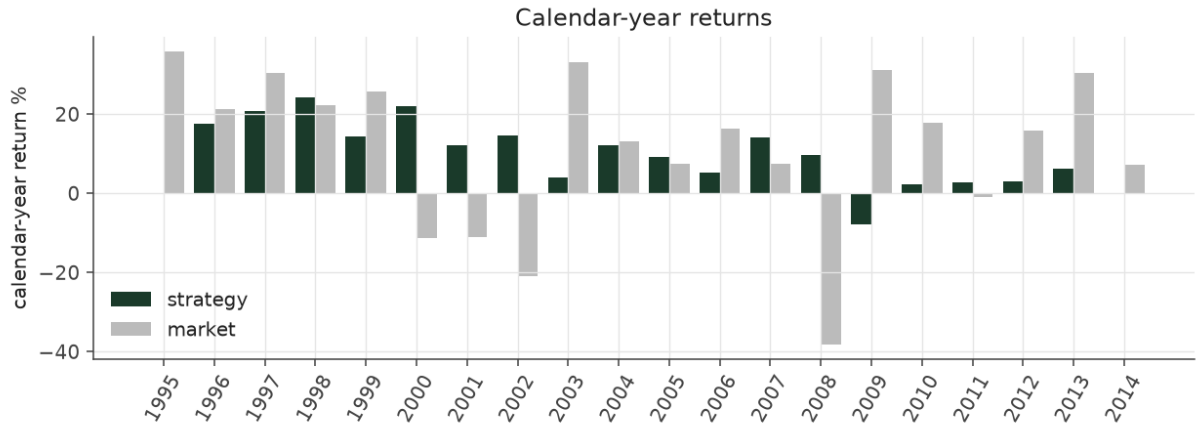
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.579



NAV (log scale)



Drawdown from running peak

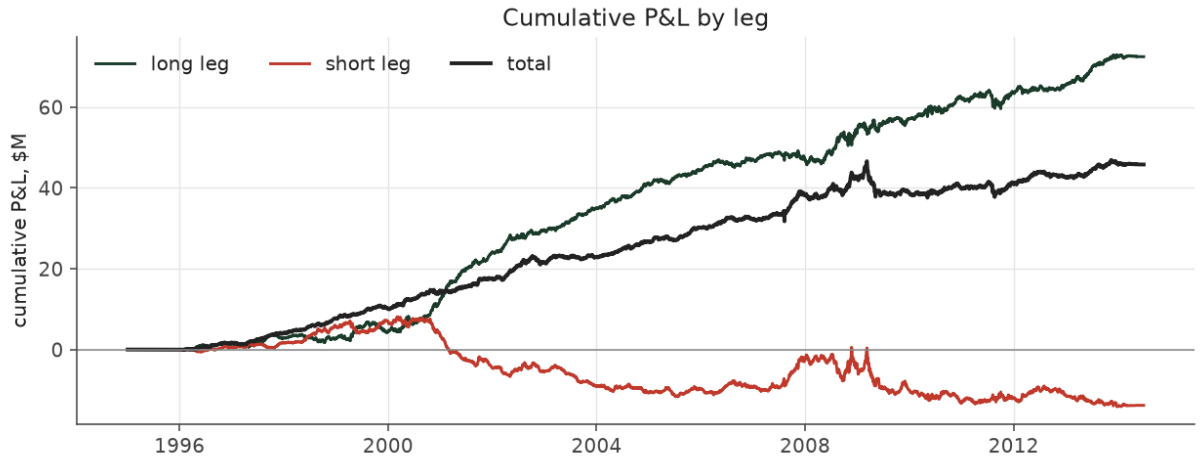


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	72.46	-13.81
P&L after costs ($M)	66.24	-20.45
of which common (average-event) component ($M)	55.02	-54.57

of which excess / PEAD component ($M)	16.80	41.19
excess minus costs ($M)	10.60	34.57
gross return per $ traded (bp)	208.91	-39.14
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	1.45	-0.16

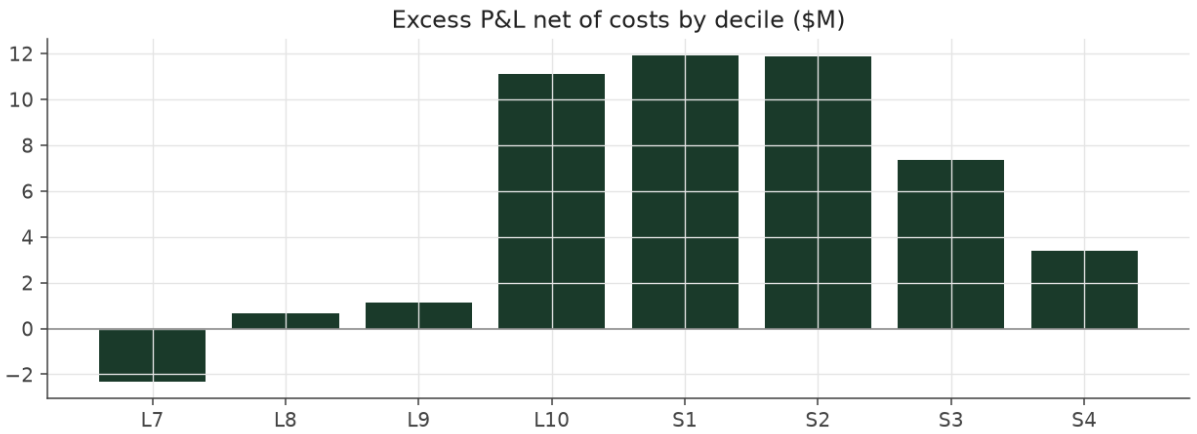
Daily P&L correlation between the two legs: **-0.77**.

• **The short leg loses money in absolute terms** (-13.8M before costs) while the long leg earns +72.5M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 55.0M / 54.6M. Stripping that common component, the surprise signal itself earned +16.8M on the long side and +41.2M on the short side.

• The two legs' daily P&L are -0.77 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.45 long / -0.16 short vs 1.73 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,029	326.55	3.79	-1.74	0.59	-2.33	-5.2%	-71.35
long	8	16,588	843.87	14.71	2.18	1.52	0.65	1.4%	7.73
long	9	17,747	1,080.59	17.10	3.04	1.91	1.14	2.5%	10.50
long	10	17,174	1,217.60	30.64	13.32	2.20	11.12	24.6%	91.34
short	1	17,169	1,171.25	-4.34	14.21	2.29	11.91	26.4%	101.73
short	2	17,281	1,084.11	-6.66	13.89	2.02	11.87	26.3%	109.49
short	3	15,970	888.83	-6.83	8.99	1.64	7.35	16.3%	82.72
short	4	9,171	384.55	-2.61	4.09	0.68	3.41	7.6%	88.70



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,213	343.64	5.44	7.77	1.29	6.48	14.3%	188.51
2	22,885	1,016.63	8.45	11.38	2.83	8.55	18.9%	84.08

3	25,501	1,544.72	12.48	14.78	3.26	11.52	25.5%	74.59
4	27,583	1,953.20	11.43	14.02	3.12	10.91	24.1%	55.84
5	28,947	2,139.15	7.99	10.04	2.37	7.67	17.0%	35.86

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,650	251.16	5.02	6.72	1.26	5.46	12.1%	217.48
2	23,084	1,138.52	8.26	13.19	3.42	9.77	21.6%	85.81
3	25,851	1,809.17	14.81	18.61	3.62	14.99	33.2%	82.84
4	27,665	1,938.19	9.63	10.59	2.71	7.87	17.4%	40.61
5	28,874	1,860.06	8.04	8.86	1.86	7.00	15.5%	37.62

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,855	2,914.53	2.37	17.73	4.88	12.85	28.4%	44.08
41-60d	25,652	1,600.35	11.07	9.46	2.99	6.47	14.3%	40.42
61-90d	35,324	1,883.42	13.71	15.05	3.65	11.40	25.2%	60.51
>90d	12,298	599.04	18.64	15.75	1.34	14.41	31.9%	240.61

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,895	1,122.39	17.52	19.48	2.06	17.42	38.6%	155.21
2002-07	39,956	2,449.43	20.47	24.05	4.43	19.62	43.4%	80.11
2008-13	45,278	3,425.53	7.80	14.45	6.37	8.08	17.9%	23.59

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	22,094	729.85	5.33	6.37	1.46	4.91	10.9%	67.27
Chems	2,590	441.60	3.43	3.99	0.69	3.30	7.3%	74.75
Durbl	2,843	460.22	2.24	3.00	0.74	2.26	5.0%	49.01
Enrgy	6,194	534.68	5.92	7.08	0.90	6.18	13.7%	115.63
Hlth	11,696	673.29	3.17	4.01	1.28	2.74	6.1%	40.63
Manuf	12,034	655.21	3.13	4.38	1.24	3.14	7.0%	47.94
Money	17,583	657.72	5.56	7.16	1.36	5.79	12.8%	88.10
NoDur	5,081	542.02	2.69	3.38	0.92	2.46	5.5%	45.43
Other	18,377	706.61	5.30	6.91	1.47	5.44	12.0%	76.99
Shops	11,984	672.08	4.85	5.88	1.28	4.60	10.2%	68.42

Telcm	3,981	466.01	2.81	3.56	0.75	2.81	6.2%	60.36
Utils	4,672	458.03	1.34	2.27	0.78	1.49	3.3%	32.57

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,523	625.15	5.22	6.76	1.11	5.64	12.5%	90.27
low vol	40,091	3,855.29	24.00	31.52	7.20	24.33	53.9%	63.10
mid vol	37,903	2,416.18	15.81	19.17	4.36	14.81	32.8%	61.32

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	9.23%	0.00%
0.5%	3.4	8.87%	0.36%
1.0%	6.9	8.49%	0.74%
2.0%	13.8	7.65%	1.58%
5.0%	34.4	3.97%	5.25%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.